Replication Plan

Spin-Dependent Scattering of Sub-GeV Dark Matter: Models and Constraints

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Auto-generated Replication Blueprint

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Contents

Paper Overview

This paper investigates spin-dependent (SD) scattering of sub-GeV dark matter (DM) off electrons in target materials, extending the DarkELF framework to handle spin-dependent interactions. The authors consider various DM models (e.g., dark photon with axial coupling, scalar mediators), compute SD scattering rates in semiconductor (Si, Ge) and molecular targets, derive exclusion limits from existing experimental data (SENSEI, DAMIC, XENON, etc.), and project sensitivity for future experiments. The calculations use DFT-derived electronic structure (wave functions and response functions).

Detailed Workflow

Phase 1: Theoretical Framework

1.1 Implement DM-electron scattering rate formulas.

- Differential scattering rate dR/dE for spin-dependent interactions.
- DM form factors for different mediator models (heavy/light mediator limits).
- DM velocity distribution (Standard Halo Model, Maxwell–Boltzmann with escape velocity).

1.2 Define target material response functions.

- Crystal form factor $|f_{\text{crystal}}(q,\omega)|^2$ from DFT band structure.
- Spin-dependent generalization of the response function.

Phase 2: Electronic Structure Inputs

2.1 Obtain DFT-computed response functions.

- Use DarkELF's built-in data for Si, Ge (from Quantum ESPRESSO DFT calculations).
- Or compute from scratch using DFT + Wannier interpolation.

2.2 Compute the dynamic structure factor $S(q,\omega)$ and its spin-dependent analog.

Phase 3: Rate Calculations

3.1 Compute scattering rates for each DM model and target material.

3.2 Sweep DM mass m_χ over the sub-GeV range (1 MeV – 1 GeV).

3.3 Compute event rates (counts/kg/year) as a function of cross-section $\bar{\sigma}_e$.

Phase 4: Experimental Constraints

4.1 Apply experimental exposure and efficiency.

- SENSEI, DAMIC, CDEX, XENON10/1T data.
- Apply energy thresholds, acceptance efficiencies, exposure.

4.2 Derive 90% C.L. exclusion limits on $\bar{\sigma}_e$ vs. m_χ.

4.3 Compute projected sensitivities for future experiments.

Phase 5: Validation

5.1 Reproduce exclusion limit plots from the paper.

5.2 Reproduce rate comparison between SI and SD scattering.

5.3 Verify against known DarkELF benchmarks for SI case.

Datasets

Dataset / Source	Type	Location / Notes
DarkELF material data (Si, Ge)	DFT response functions	Bundled with DarkELF `https://github.com/tongylin/DarkELF`
Experimental exposure data	Published	SENSEI, DAMIC, XENON papers
DM halo parameters	Standard	$v_0 = 220\,\mathrm{km/s}$, $v_{\mathrm{esc}} = 544\,\mathrm{km/s}$, $\rho_\chi = 0.4\,\mathrm{GeV/cm^3}$
Mediator model parameters	From paper	Coupling constants, mediator masses

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
DarkELF	latest	DM-electron scattering rate calculator `https://github.com/tongylin/DarkELF`
Python	≥ 3.8	Primary language
NumPy / SciPy	≥ 1.21	Numerical integration, interpolation
Matplotlib	≥ 3.4	Exclusion limit and rate plots
Quantum ESPRESSO	latest	DFT calculations (if recomputing material data) `https://www.quantum-espresso.org/`
QEdark	latest	Alternative DM-electron code `https://github.com/adrian-soto/QEdark`
Jupyter	latest	Interactive analysis

Hardware Requirements

- Rate calculations with DarkELF: any laptop (seconds to minutes per mass point).
- DFT calculations (if needed): HPC cluster, ≥ 32 cores, $\geq 64\,\mathrm{GB}$ RAM.